

The Welfare Cost of Anticipated Inflation in the United Kingdom

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Abstract

This dissertation estimates the steady-state welfare cost of anticipated inflation in the United Kingdom using the Bailey–Lucas consumer-surplus mapping from money demand. Using annual data for 1980–2024—M1 (end-December), nominal GDP, and the Bank Rate—I model the money–income ratio $z = M_1/Y$ as a function of the user cost i under two standard forms: a constant-elasticity log–log and a semi-log. The preferred log–log specification (interest elasticity $g \approx -0.054$) implies very small losses at policy-range user costs: lowering i from 2% to 0% yields a welfare gain of about 0.004% of income. The semi-log mapping is flatter near $i=0$ and delivers even smaller numbers ($\sim 0.0007\%$ at 2%). Interpreted through the Friedman rule, the UK’s low-inflation regime already leaves only a tiny wedge between private and social liquidity costs; a small positive target remains defensible on stabilisation and second-best fiscal grounds. I discuss measurement limits (aggregate choice, user-cost proxy, structural breaks) and note how broader GE frameworks (MIU/CIA/search) could shift levels without overturning the central finding: in modern UK institutions, the steady-state welfare cost of low, stable inflation is modest.

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1 Introduction

Anticipated inflation, even when fully priced into contracts, imposes welfare costs by raising the private user cost of liquidity and inducing households and firms to economise on real money balances. In the Bailey–Lucas framework, these costs appear as the area under a money–demand curve between the prevailing nominal user cost of money and the zero–cost benchmark. The United Kingdom provides a particularly informative setting for measuring these costs: the policy regime transitioned from disinflation in the early 1980s, to inflation targeting in the 1990s, to the effective lower bound after 2008, and back to rapid tightening in 2021–2024. This variation in the nominal user cost i offers leverage for estimating the shape of transactions money demand and for mapping estimated schedules into steady-state welfare costs of anticipated inflation.

This dissertation measures those costs for the UK using annual data from 1980–2024 and a deliberately transparent methodology. I model the money–income ratio $z = M/Y$ —where M is an M1-style transactions aggregate and Y is nominal income—as a function of i under two workhorse forms: (i) a constant-elasticity (log–log) specification and (ii) a semi-log specification. Under unit income elasticity and standard stock–flow timing, the welfare cost of a steady nominal user cost i is read directly from the money–demand schedule. The estimates imply that, at low policy-relevant user costs, the deadweight loss (as a *share of income*) is small—a finding consistent with international evidence for advanced economies and informative about the role of a low positive inflation target in modern UK institutions. A dedicated subsection in Section 4 situates the UK magnitudes within that literature.

1.1 Motivation and contribution

The motivation is twofold. First, the inflation-targeting regime explicitly trades off stabilisation goals against long-run efficiency considerations. Quantifying steady-state welfare costs at different user-cost levels clarifies what is at stake when policymakers choose targets and the pace of disinflation. Second, the UK has experienced pronounced changes in payments technology, bank regulation, and interest-rate environments over the sample. These institutional shifts are precisely the kind of forces that can bend the money–demand schedule, particularly in the neighbourhood of $i \rightarrow 0$, where functional-form assumptions matter most.

I contribute by bringing up-to-date UK evidence to a classic question with three elements of value added. *First*, I assemble an internally consistent dataset aligned with the Bailey–Lucas mapping: an end-period transactions aggregate for M , flow income Y , and a policy-rate user-cost proxy i . *Second*, I estimate both log–log and semi-log demand in levels and report closed-form welfare expressions, highlighting how curvature near the

lower bound drives the resulting losses. *Third*, I interpret magnitudes through the lens of the Friedman rule and second-best considerations, and explicitly benchmark the UK against other advanced and emerging economies (Section 4.4), explaining why the empirical finding of “small but non-zero” welfare costs aligns with a positive, low inflation target in practice.

1.2 Conceptual framework

Let $z(i) \equiv M(i)/Y$ denote desired real transactions balances per unit of nominal income as a function of the nominal user cost of money i . In the Bailey–Lucas consumer–surplus approach (Bailey, 1956; Lucas, 2000), the steady-state welfare loss from a constant $i > 0$ is the *net* area between the money–demand schedule and the price line:

$$W(i) = \int_0^i z(x) dx - i z(i),$$

measured as a share of income. I consider two tractable forms:

1. **Log–log (constant elasticity).** $\log z = \alpha + g \log i$ with $g < 0$. Then $z(i) = e^\alpha i^g$ and

$$W(i) = \frac{-g}{1+g} e^\alpha i^{1+g} \quad (g \in (-1, 0)).$$

2. **Semi-log (semi-elasticity).** $\log z = \beta + \gamma i$ with $\gamma < 0$. Writing $\varepsilon \equiv -\gamma > 0$, we have $z(i) = e^\beta e^{-\varepsilon i}$ and

$$W(i) = \frac{e^\beta}{\varepsilon} \left[1 - (1 + \varepsilon i) e^{-\varepsilon i} \right].$$

These expressions make transparent how the *level* of z and its *curvature near* $i = 0$ determine welfare. The two forms bracket behaviour at low user costs, which is where the UK data concentrate.

1.3 UK institutional background

The UK underwent major monetary and financial transformations over the sample horizon. Inflation targeting was formalised in the early 1990s, the operational independence of the Bank of England was granted in 1997, and the policy framework weathered the global financial crisis, the effective lower bound (ELB) era, and the inflation surge after 2021. In parallel, retail payments technology (contactless, Faster Payments, mobile wallets) and cash-substitute instruments expanded, while the composition and interest-sensitivity of deposits evolved. These features matter for money demand: greater availability of low-friction substitutes for cash-like balances tends to flatten the demand curve with respect to i , lowering the implied welfare triangle for a given user cost.

From a measurement perspective, the choice of aggregate is consequential. A narrow M1-style measure targets transactions balances most closely but is affected by institutional reclassification and financial innovation. Broader aggregates include assets with different opportunity costs and liquidity services. My baseline adopts a consistent transactions-oriented stock and scales by nominal income to form z , recognising that this choice trades potential measurement error against conceptual alignment with the Bailey–Lucas mapping.

1.4 Estimation strategy

I estimate money demand in levels using ordinary least squares for both forms:

$$\begin{aligned}\log z_t &= \alpha + g \log i_t + u_t, \\ \log z_t &= \beta + \gamma i_t + u_t \quad (\varepsilon \equiv -\gamma > 0),\end{aligned}$$

where t indexes years. Under the log–log form, the interest elasticity g directly governs the curvature of the welfare schedule. Under the semi-log form, the semi-elasticity ε controls how rapidly money demand decays with i . I then map the fitted parameters into the closed-form welfare expressions above and evaluate $W(i)$ at representative user-cost levels. Because z is persistent and i is policy-driven, I examine residual diagnostics and sensitivity to excluding extreme years (e.g., crisis or pandemic) to guard against spurious fit or regime-driven outliers. The data definitions and construction are detailed in Section 3; estimation results, welfare maps, and the cross-country placement of UK magnitudes are reported in Section 4 (see Section 4.4).

1.5 Mapping estimates to welfare and the Friedman rule

The Friedman rule prescribes $i = 0$ in a frictionless environment where money can be provided at negligible social cost, eliminating the wedge between private and social valuations of liquidity. My mapping measures the cost of deviating from this benchmark using $W(i) = \int_0^i z(x) dx - i z(i)$. For a given parameterisation, the welfare gain from moving from $i = i_1$ to $i = i_0 < i_1$ is $\Delta W = \int_{i_0}^{i_1} z(x) dx$. When i is already small, the concavity of $W(i)$ implies diminishing gains from further disinflation; conversely, keeping i elevated generates rapidly increasing losses. Interpreting the UK’s low-inflation regime through this lens, the efficiency benefits of shaving a few additional basis points from an already low i are modest, whereas avoiding prolonged periods of high i yields sizable welfare protection.

Crucially, the Friedman benchmark is a *first-best* result. In practice, central banks weigh stabilisation benefits, the risk of binding lower bounds, and seigniorage considerations. My partial-equilibrium welfare metric does not capture these general-equilibrium

trade-offs directly, but it quantifies the steady-state triangle one would be willing to forgo to reduce i . The small empirical magnitudes at low i support the case for a positive but low target: efficiency losses are tiny, while macro-stabilisation flexibility and lower-bound insurance may be valuable.

1.6 Preview of results

The constant-elasticity (log–log) demand curve is flat in the policy-relevant range, implying small triangles at modest i . For illustration, reducing i from 2% to 0% yields a welfare gain of about 0.004% of income under the log–log form, and roughly 0.0007% under the semi-log form. At much higher user costs (e.g. 8–10%), the losses scale up several-fold but remain a small share of income by macroeconomic standards. These magnitudes match the visual curvature of the fitted schedules and are consistent with international evidence for advanced economies; Section 4.4 places the UK explicitly within that distribution.

1.7 Relation to existing literature

Classical treatments quantify the welfare cost of inflation by integrating under money-demand schedules (the Bailey triangle) or by embedding money in preferences or constraints (cash-in-advance, money-in-utility, shopping-time). In advanced economies, partial-equilibrium estimates typically find small deadweight losses at low, stable inflation, while environments where liquidity is essential for decentralised exchange can imply larger costs when frictions bind. Empirical applications stress the importance of aggregate definition, user-cost measurement, and sample period. My contribution is to provide a unified, UK-focused update using contemporary data and to report side-by-side results under the two canonical functional forms, thereby clarifying how near-zero behaviour of demand shapes the welfare mapping and how the UK compares internationally (Section 4.4).

1.8 Scope, limitations, and robustness

The approach is intentionally narrow to ensure transparency. It abstracts from heterogeneity in money demand across households and firms, from precautionary motives linked to risk, and from general-equilibrium feedbacks between monetary and fiscal policy. Measurement choices—the use of a policy-rate proxy for the user cost, the reliance on a narrow transactions aggregate, and the stock–flow timing—introduce potential biases. Structural breaks associated with financial liberalisation, the adoption of inflation targeting, the ELB, and the pandemic may alter elasticities over time. I address these concerns by (i) comparing log–log and semi-log forms, (ii) examining residual diagnostics and excluding extreme years in sensitivity checks, and (iii) interpreting magnitudes as steady-state benchmarks rather than precise short-run policy prescriptions.

1.9 Policy relevance

The welfare mapping directly informs normative discussions about target design. When user costs are near zero, the measured efficiency loss from a small positive i is tiny, strengthening the case for targets that prioritise stabilisation, financial stability, and lower-bound insurance. When user costs are elevated for prolonged periods, the welfare triangle grows quickly, highlighting the importance of credibility, timely disinflation, and avoiding persistent drift in inflation expectations. The UK estimates thus provide both a calibration point for the long-run trade-off and a metric to evaluate the welfare implications of different paths of policy normalisation.

1.10 Structure of the dissertation

The remainder proceeds as follows. Section 2 surveys the related literature on money demand and welfare costs of inflation. Section 3 documents data construction, variable definitions, and UK institutional features relevant for measuring transactions balances and user costs. Section 4 sets out the empirical methodology, reports the money-demand estimates, maps them into welfare, and benchmarks the UK in an international context (Section 4.4). Section 5 concludes with policy implications and avenues for future research.

2 Literature Survey

2.1 Approaches to Measuring the Welfare Cost of Inflation

The existing literature employs several distinct methodological frameworks to quantify the welfare cost of anticipated inflation. Each approach examines a particular channel through which inflation distorts economic behaviour, resulting in welfare cost estimates that vary considerably depending on the underlying modelling assumptions, the definition of money adopted, and institutional characteristics such as payments technology and the availability of interest-bearing transaction accounts.

(i) Consumer surplus (Bailey–Lucas) approach. This approach treats real money balances as providing liquidity services, with the nominal interest rate serving as the user cost of these services. The welfare loss represents the deadweight loss arising from agents holding fewer real balances than would be optimal under zero nominal interest rates, as inferred from empirically estimated money demand relationships. The theoretical foundations trace back to Bailey (1956), who developed the analytical framework, while Cagan (1956) provided early empirical applications focusing on money demand behaviour in hyperinflationary environments. This methodology's empirical implementation and cross-country evidence are examined in detail below.

(ii) Money-in-utility (MIU) models. MIU specifications incorporate real money balances directly into household utility functions, such that inflation reduces welfare by diminishing the return to liquidity holdings and compelling households to adjust their saving and expenditure decisions. This approach provides microfoundations for money demand while requiring careful calibration of preference parameters.

(iii) Cash-in-advance (CIA) constraint models. CIA frameworks impose timing restrictions requiring certain purchases to be financed through cash holdings. Elevated nominal interest rates tighten these constraints and generate general equilibrium distortions affecting consumption, labour supply, and investment decisions. By design, these models capture portfolio and timing responses beyond the scope of partial equilibrium consumer surplus measures.

(iv) Shopping-time and inventory-theoretic models. This class of models recognises that money economises on real resources required for payment processing, including time, transportation costs, and other transaction inputs. Inflation increases the opportunity cost of holding idle balances relative to expending resources on payment activities, thereby generating genuine "shoe-leather" costs. The approach provides an intuitive

framework for understanding the resource costs of cash management in different institutional environments.

(v) Search-theoretic (decentralised exchange) models. Search-based models of monetary exchange treat money as essential because anonymous buyers and sellers encounter each other bilaterally and cannot establish enforceable credit arrangements. Building upon the Kiyotaki–Wright tradition and its tractable formalisation by Lagos and Wright (2005), these models assume agents select cash holdings prior to random matching opportunities. Inflation erodes real balances, causing buyers to enter meetings with reduced liquidity, which consequently (i) constrains feasible transaction volumes, (ii) deteriorates the buyer’s terms of trade under bargaining or price-posting mechanisms, and (iii) may deter matches entirely. Welfare calculations compare steady-state consumption-equivalent utility across alternative inflation regimes, capturing both liquidity effects and distorted surplus distribution—channels that consumer surplus, MIU, and CIA approaches typically abstract from. While the Friedman rule (zero nominal return on money) often emerges as optimal, departures from this prescription can be justified under certain conditions involving bargaining power or search externalities (Craig & Rocheteau, 2006).

2.1.1 Consumer Surplus and the Money Demand Function

The systematic investigation of inflation’s welfare consequences originates with Bailey (1956), who conceptualises the liquidity services provided by money as a standard consumption good, with the nominal interest rate representing its user cost. When inflation drives up nominal rates, private agents reduce their real balance holdings (maintaining fewer transaction balances while adopting more sophisticated cash management strategies), and the resulting foregone liquidity services constitute a deadweight loss. Bailey’s public finance framework suggests that this loss can be measured directly from observed money demand behaviour: it equals the net reduction in consumer surplus when the "price" of liquidity increases from zero to the prevailing nominal rate. Since newly created money generates seigniorage revenue for the government, this revenue component must be subtracted to obtain the true social cost.

Bailey’s insight transformed the welfare cost question into an empirical exercise: estimate a money demand schedule and integrate beneath it. Pursuing this research agenda, Lucas (2000) demonstrates that functional form assumptions for money demand prove crucial at low nominal interest rates. Using U.S. data spanning 1900–1994, he contrasts a constant-elasticity (log-log) specification with a semi-log specification traditionally employed in high-inflation contexts such as Cagan (1956). Lucas reports that reducing steady-state inflation from 10% to 0% would increase measured income by *slightly less than* one percent, with the constant-elasticity specification providing superior fit to the extended U.S. sample. He emphasises that much of the variation across studies stems

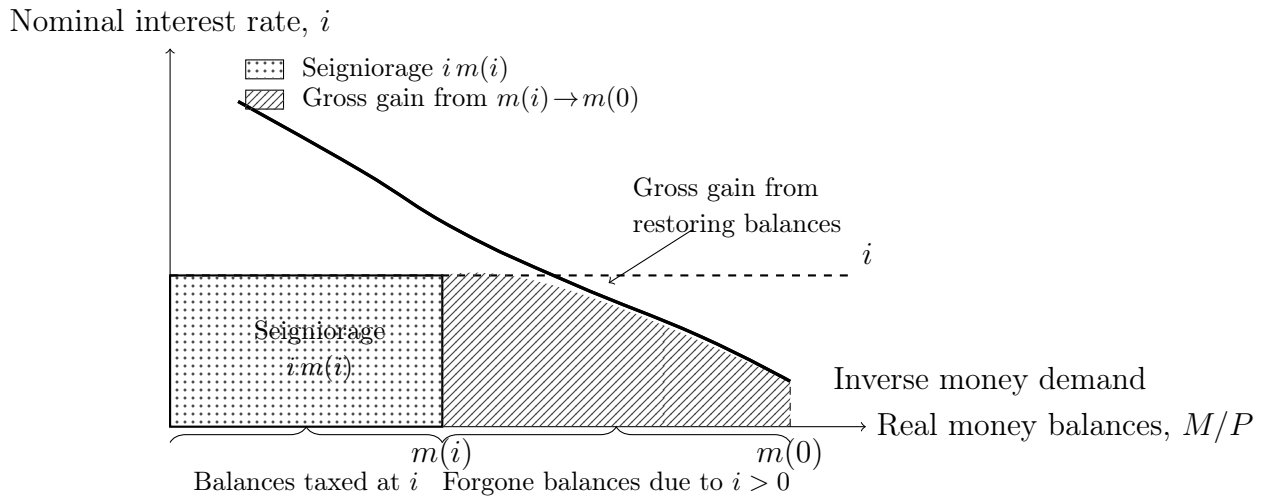


Figure 1: Bailey consumer-surplus decomposition. The dashed line marks the user cost i . When $i > 0$, agents choose $m(i)$ rather than $m(0)$, reducing liquidity services. The *dotted rectangle* shows seigniorage $i m(i)$; the *hatched area* shows the gross gain from restoring balances to $m(0)$. The net welfare cost of anticipated inflation equals the hatched area minus the dotted rectangle.

from behaviour near zero nominal rates, where the alternative demand curve specifications imply markedly different areas under the curve.

Two methodological considerations repeatedly surface in this literature. First, the selection of monetary aggregate and opportunity cost proxy exercises decisive influence: employing the monetary base versus M1, or a Treasury bill rate versus a deposit rate, alters both the level and interest sensitivity of estimated money demand. Second, financial innovation shifts money demand relationships over time, so estimates from earlier periods may not characterise modern behaviour. Consequently, studies re-estimating money demand using post-1980 data often uncover substantially smaller consumer surplus costs at policy-relevant inflation rates compared to exercises employing longer historical samples, even when applying identical methodological frameworks (Ireland, 2009). Conceptually, however, the Bailey approach retains its status as the benchmark partial equilibrium measure of steady-state anticipated inflation's tax on liquidity services, yielding results that remain transparent and comparable across different institutional settings once definitional harmonisation is achieved.

2.1.2 Friedman's Optimum Quantity of Money

Friedman (1969) establishes the normative benchmark against which these empirical measures should be evaluated. Given that base money can be produced at approximately zero marginal resource cost, allocative efficiency demands elimination of the private wedge that encourages economising on liquidity. Friedman's prescription calls for policy to engineer conditions under which holding money becomes costless at the margin—equivalently,

driving the nominal interest rate to approximately zero—thereby ensuring private agents achieve satiation in real balances. Within a frictionless environment, this outcome can be accomplished by setting steady-state deflation equal to the real interest rate, such that money’s real return matches that available on safe bonds. Under such alignment, no incentive exists to divert real resources toward cash management devices, and the Bailey consumer surplus area disappears.

The Friedman rule serves as a powerful organising principle by identifying the first-best benchmark against which empirical welfare losses from positive trend inflation should be measured. Nevertheless, several considerations explain why contemporary central banks typically target low but positive inflation rather than implementing the rule literally. First, reducing the nominal rate to zero during normal economic conditions may constrain policymakers at the effective lower bound during recessions; maintaining a modest positive target preserves scope for rate reductions. Second, governments depend on seigniorage alongside distortionary taxation; absent lump-sum instruments, optimal policy must balance the inflation tax against other tax wedges, so the rule need not represent the second-best optimum in practice. Third, within environments characterised by nominal rigidities or distributional and liquidity frictions beyond the transaction margin, slightly positive inflation may help mitigate distortions that the simplified benchmark ignores. In summary, Friedman (1969) provides the theoretical efficiency criterion—eliminate liquidity’s opportunity cost—while empirical research such as Lucas (2000) and subsequent refinements quantify how substantially actual regimes deviate from this ideal in income-equivalent terms.

2.2 Evidence by Country Group

Monetary institutions and payments technology fundamentally shape the severity of steady-state inflation’s welfare burden. In advanced economies characterised by widespread interest-bearing transaction balances and sophisticated electronic payment systems, most consumer surplus, MIU, and CIA estimates place policy-relevant inflation’s welfare costs firmly within the "modest" category. When liquidity plays an essential role in decentralised trading relationships, estimated losses may prove larger. Conversely, in emerging and developing economies where cash dependence remains high and close substitutes for transaction balances remain scarce, identical methodological approaches typically generate substantially higher welfare cost estimates.

2.2.1 Advanced Economies (United States, OECD, and Other High-Income Countries)

United States (consumer surplus approach). The extensive U.S. historical record provides the benchmark for this literature. Analysing data from 1900–1994, Lucas (2000)

estimates that reducing steady-state inflation from 10% to zero would yield benefits worth *slightly less than* 1% of income, with the constant-elasticity specification providing superior fit to the extended sample. He emphasises that much of the variation across studies stems from behaviour near zero nominal rates, where alternative demand curve specifications imply markedly different areas under the curve. When Ireland (2009) re-examines the post-1980 period using a semi-log money demand specification, the estimated costs at policy-relevant rates prove much smaller: approximately **0.036–0.038%** of income at 2% inflation and **0.224–0.232%** at 10%. This contrast effectively illustrates how sample period selection and functional form assumptions critically influence consumer surplus magnitudes.

United States (general equilibrium analysis). General equilibrium models corroborate the "small but non-zero" characterisation for moderate inflation. Within a benchmark real business cycle framework incorporating CIA constraints, sustained 10% inflation imposes costs of approximately **0.11–0.40%** of GNP (Cooley & Hansen, 1989). By construction, these models capture portfolio reallocation and timing responses that exceed the scope of partial equilibrium consumer surplus measures. Search-theoretic models, as discussed above, can yield substantially higher estimates when liquidity plays a central matching role.

Advanced economy microeconomic evidence (shopping-time approach). Household-level behavioural analysis supports the post-1980 consumer surplus findings. Exploiting comprehensive Italian microdata on cash holdings, shopping trips, ATM adoption, and withdrawal patterns, Attanasio et al. (2002) conclude that resource costs of cash management within the policy-relevant range remain *modest*—approximately **0.1% of consumption or less**. Low fixed rebalancing costs in modern payment environments help explain these small magnitudes.

OECD cross-country patterns (search-based analysis). Even among high-income economies, institutional details matter substantially. Implementing a unified Lagos–Wright framework across 23 OECD members (1978–1998), Boel and Camera (2011) document considerable dispersion: under price-taking behaviour, many countries would sacrifice only a fraction of 1% of consumption to avoid 10% inflation (Japan approaches **0.1%**), while others register higher costs (Australia exceeds **2%**); incorporating bargaining frictions drives some estimates toward **6%**. Payments technology sophistication and interest-bearing transaction balance availability align with this pattern.

Canada versus United States (long-horizon consumer surplus). Long-run elasticity identification introduces an additional refinement. Employing Fisher–Seater long-

horizon methods within a log-log money demand framework (1948–2001), Serletis and Yavari (2004) obtain smaller consumer surplus losses than classic U.S. long-sample benchmarks: for the United States, reducing nominal rates from 14% to 3% yields gains worth approximately **0.45%** of income, while a further reduction from 3% to 0% adds **0.18%**; for Canada, corresponding gains equal **0.35%** and **0.15%** respectively. The key insight is that careful long-run identification tends to compress consumer surplus estimates.

Collectively, evidence from advanced economies employing modern payment systems suggests broad methodological agreement: low trend inflation imposes modest steady-state costs, while models that assign central importance to liquidity within matching frameworks can imply larger benefits from achieving complete price stability.

2.2.2 Emerging and Developing Economies

China (consumer surplus combined with MIU and CIA approaches). Where cash maintains greater centrality and substitutes remain limited, measured welfare losses increase substantially. Analysing Chinese data from 1996–2006, Chen and Ma (2007) compute Bailey consumer surplus measures under both log-log and semi-log money demand specifications, complemented by MIU and CIA calibrations. The consumer surplus estimates prove large: at **10%** inflation, costs reach **11.25%** of consumption under log-log demand (with **9.41%** at 8% and **6.46%** at 5%); under semi-log demand, 10% inflation costs **9.69%**. The CIA calibration yields substantially lower general equilibrium losses—**0.912%** of consumption at 10%. Year-by-year calculations support this pattern: in **1996**, estimated losses reached approximately **12.99%** of consumption (roughly **4.38%** of GDP; approximately ¥297.15 billion), declining by **2005** to approximately **0.87%** of GDP (approximately ¥159.17 billion).

Brazil. Post-Real Plan estimates employing long-run money demand methodologies yield sub-percent costs within policy ranges: approximately **0.15–0.20%** of income at 4.5% inflation and **0.3–0.45%** at 10%. Allowing the elasticity to vary over time through cointegration and Kalman filtering techniques improves statistical fit while generating similar magnitudes for given inflation rates (Campos & Cysne, 2019).

South Africa. Methodological selection proves decisive. Using Johansen cointegration (M3, short safe rate), consumer surplus costs across the **3–6%** target band range from **0.34–0.67%** of GDP; identifying long-run elasticity through Fisher–Seater long-horizon regressions approximately halves these estimates to **0.16–0.36%** (Gupta & Uwilingiye, 2008, 2009).

Turkey. For the 1970–2013 period, log-log demand specifications prove preferable; increasing steady-state inflation from **0%** to **10%** implies consumer surplus losses of **0.52–0.54%**

of GDP using M1 (Tümtürk, 2017).

India. Combining consumer surplus and compensating variation (general equilibrium) measures over 1996–2015, Shah et al. (2019) estimate that **10%** inflation costs approximately **0.53%** of GDP, with magnitudes sensitive to money demand functional form selection.

Colombia. Within a Sidrauski-type MIU calibration, welfare costs reach **1.37%** of GNP at **10%** inflation and approximately **2.4%** of GDP at **20%** (López, 2000).

Collectively, emerging market evidence demonstrates that identical conceptual frameworks can generate substantially larger welfare cost estimates when transaction technologies remain cash-intensive and payment systems less developed. The variation across studies primarily reflects three modelling decisions: the monetary aggregate and user cost proxy selection, the functional form adopted for money demand (log-log versus semi-log), and the approach used for long-run elasticity identification (cointegration versus long-horizon versus time-varying methods).

2.3 Literature Gaps and Outstanding Questions

The evidence surveyed above indicates that welfare costs at policy-relevant inflation rates typically remain modest in advanced economies when measured through consumer surplus or CIA/shopping-time methodologies, though estimates can prove larger within search-theoretic frameworks and in contexts where cash plays a central role in exchange. Several gaps persist concerning the UK context:

(i) Limited UK-specific quantification. Despite extensive U.S. and OECD evidence, remarkably little research has estimated UK welfare costs using contemporary data and comparative methodological approaches. Given that national institutional characteristics influence money demand behaviour and the opportunity cost of balance holdings, UK-specific estimates represent a necessary contribution rather than merely applying U.S. benchmarks.

(ii) Contemporary institutional environment. The UK's inflation-targeting framework and post-2008 payments innovations (including widespread electronic payment adoption and interest-bearing current accounts) have plausibly altered the income elasticity, interest semi-elasticity, and user cost confronting households and firms. Re-estimating money demand relationships across different regime periods could illuminate how contemporary institutional arrangements translate into welfare implications.

(iii) Functional form sensitivity at low interest rates. The divergence between log-log and semi-log money demand specifications proves most pronounced at low nominal rates. UK estimates should transparently report both consumer surplus curves and quantify how functional form selection influences welfare calculations near the effective lower bound.

(iv) Opportunity cost measurement. Relying upon a single policy rate (Bank Rate) may mischaracterise the user cost of transaction balances when retail deposit rates diverge or when interest-bearing transaction accounts become prevalent. Robustness analysis across alternative user cost proxies and money aggregates (narrow versus broader transaction measures) remains essential.

(v) Methodological comprehensiveness. Consumer surplus measures offer transparency and straightforward data mapping; MIU and CIA models incorporate portfolio and timing dimensions; search-theoretic models value liquidity within decentralised trading arrangements and can imply larger losses (Boel & Camera, 2011). A UK study reporting comparable results across these complementary approaches would prove particularly informative.

This dissertation's contribution. To address these identified gaps, the subsequent empirical analysis (i) estimates UK money demand over 1980–2024 under both log-log and semi-log specifications (assuming unit income elasticity); (ii) computes welfare costs via the consumer surplus approach; and (iii) benchmarks the resulting UK estimates against U.S./OECD and emerging market evidence.

3 Data

This section documents the construction of the annual UK dataset employed in the welfare cost estimation. The variables, sources, and transformations align with the Bailey–Lucas consumer surplus framework under unit income elasticity assumptions. The sample spans **1980–2024** and follows calendar year conventions: M1 represents an end-of-year stock, nominal GDP an annual flow, and Bank Rate a calendar-year average. The core empirical object is the money–income ratio $z_t = M_{1,t}/Y_t$, paired with the nominal user cost i_t .

3.1 Variables, Sources, and Empirical Mapping

The dataset construction follows the Bailey-Lucas consumer surplus framework under unit income elasticity assumptions. Three annual UK series enter the analysis over **1980–2024**:

- **Transaction balances** $M_{1,t}$: M1 (notes and coin plus sight deposits), end-December stock, £ billions (Bank of England).
- **Nominal activity** Y_t : nominal GDP at market prices, £ billions (ONS, UK Economic Accounts, series YBHA).
- **User cost** i_t : Bank Rate, calendar-year average, decimal (Bank of England).

Under unit income elasticity assumptions, the empirical object becomes the money–income ratio $z_t \equiv M_{1,t}/Y_t$, which maintains proportionality to real balances per unit of real output and therefore eliminates the need for explicit price deflation. The analysis relates z_t to i_t to recover a long-run money demand schedule and maps this relationship into welfare costs via standard consumer surplus calculations (see Section 4).

Data construction procedures.

1. *Temporal aggregation*: Monthly M1 data aggregated to annual end-December stocks; daily Bank Rate data averaged to calendar-year observations; GDP data obtained as published annual figures.
2. *Units*: $M_{1,t}$ and Y_t expressed in £ billions; i_t expressed in decimals.
3. *Derived variables*: $z_t = M_{1,t}/Y_t$; logarithmic transformations $\log z_t$ and $\log i_t$ computed directly.
4. *Sample coverage*: Complete annual coverage for 1980–2024; no missing observations.

3.2 Summary Statistics and Empirical Patterns

This subsection characterises the primary data series and establishes motivation for the empirical specifications employed below. Three empirical regularities emerge over the 1980–2024 period. First, both M1 and nominal GDP exhibit strong nominal growth, consistent with long-run economic expansion and the increasing prominence of sight deposits within a highly intermediated financial system. Second, Bank Rate demonstrates substantial variation across monetary policy regimes—elevated during the early 1980s disinflation, constrained at the effective lower bound following 2008, and rising again during the early 2020s—providing the variation necessary to identify interest sensitivity in money demand. Third, the money-income ratio $z = M_1/Y$ displays persistence while remaining comparatively modest by advanced economy standards, consistent with the ease of rebalancing between non-interest-bearing and interest-bearing transaction instruments in the UK financial system.

Table 1: Summary statistics (annual, 1980–2024)

Variable	Mean	Median	SD	Min	Max	N
M1 (£bn)	45.894	39.439	28.628	11.721	98.756	45
Nominal GDP (£bn)	1280.726	1191.439	708.037	259.970	2850.989	45
Bank Rate (decimal)	0.057	0.051	0.046	0.001	0.163	45
$z = M_1/Y$	0.035	0.035	0.005	0.028	0.045	45

Component series analysis. The individual trajectories of M1 and nominal GDP illuminate the underlying drivers of movements in $z = M_1/Y$. Figure 2 documents the steady expansion of M1 balances—initially gradual, subsequently accelerating through the 2000s and early 2010s—consistent with financial deepening and the proliferation of electronic payment systems. Figure 3 presents nominal GDP growth that intensifies during high-inflation periods; when nominal GDP expansion exceeds M1 growth, the ratio z contracts accordingly. The interaction between these trajectories anticipates the z dynamics presented in Figure 4.

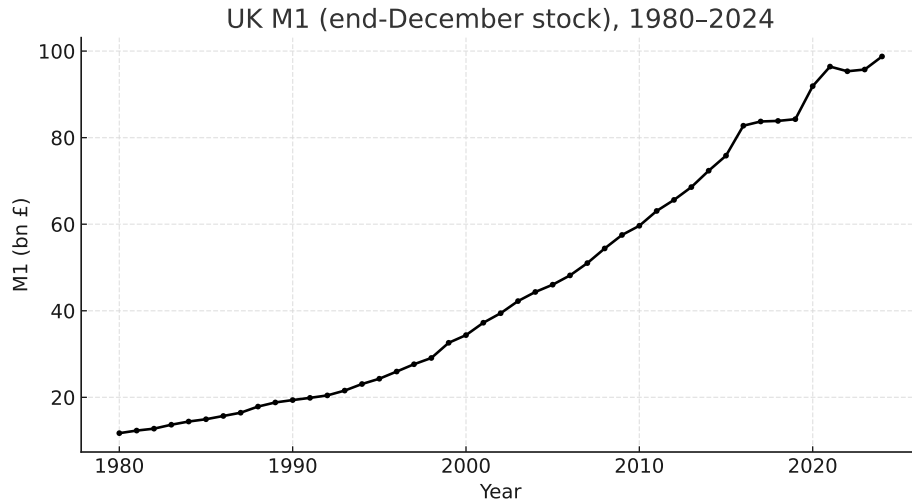


Figure 2: UK M1, end-December stock (£bn), 1980–2024. *Notes:* Expansion reflects financial deepening and the growing importance of sight deposits within retail financial intermediation.

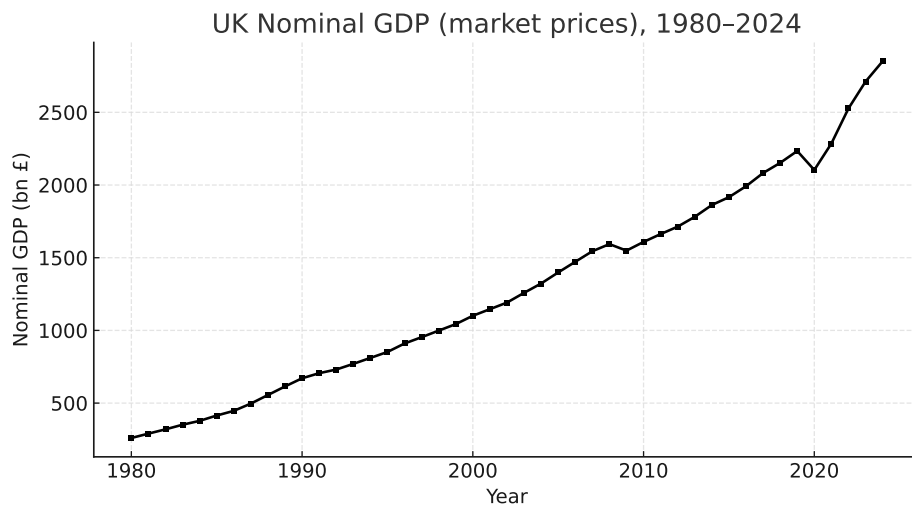


Figure 3: UK nominal GDP at market prices (£bn), 1980–2024. *Notes:* Growth accelerates during high-inflation episodes; faster nominal GDP growth relative to M1 mechanically compresses $z = M_1/Y$.

The money-income ratio evolution. Figure 4 presents $z = M_1/Y$ over the sample period. The ratio declines throughout the 1980s, reaches a trough during the early 1990s, exhibits an upward trend through the 2000s and early 2010s, spikes around 2020, and subsequently falls as nominal GDP accelerates during the 2021–2024 inflation episode. This pattern aligns closely with UK institutional developments and monetary policy regimes: disinflation and elevated user costs during the 1980s; financial deepening and electronic payment diffusion supporting higher sight deposit holdings through the 2000s; precautionary liquidity accumulation during the pandemic; and subsequent compression of z as price levels rose rapidly.

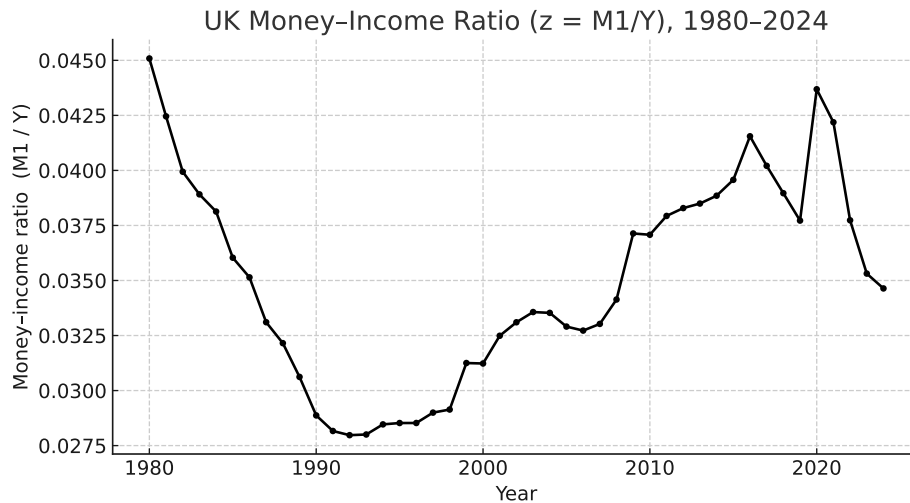


Figure 4: Money-income ratio $z = M_1/Y$, annual (1980–2024). *Notes:* Pronounced decline during the 1980s; trough in early 1990s; steady increase through 2000s–early 2010s; pandemic-era spike; subsequent decline as nominal GDP accelerates post-2021. The series exhibits persistence, consistent with a stable long-run money demand relationship.

User cost variation across monetary regimes. Figure 5 plots Bank Rate (annual averages) across the sample period. The temporal pattern reflects major UK monetary policy regimes: elevated rates during early 1980s disinflation; ERM exit and transition to inflation targeting during the 1990s; operational independence from 1997; post-2008 effective lower bound constraints; and the 2021–2024 tightening cycle. Observations near the effective lower bound prove particularly informative for distinguishing functional form specifications, since behaviour as $i \rightarrow 0$ critically determines differences between log-log and semi-log money demand relationships.

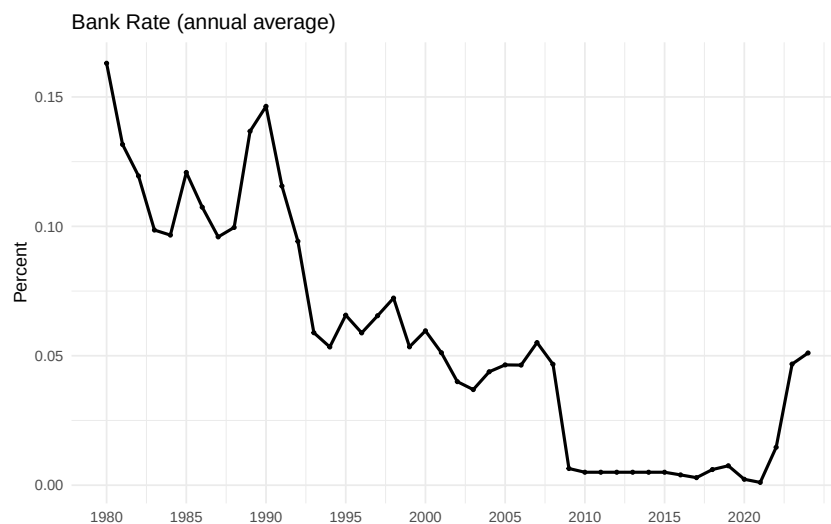


Figure 5: Bank Rate, percent (annual average, 1980–2024). *Notes:* Substantial regime variation—effective lower bound post-2008 and sharp increases during early 2020s—provides leverage to identify interest sensitivity across both low and moderate user cost ranges.

Identification implications. The visual evidence suggests (i) high persistence in z and (ii) substantial variation in i across policy regimes. Within an institutional environment characterised by widespread interest-bearing current accounts and sophisticated electronic payment systems, modest interest elasticity at policy-relevant rates appears plausible and, as demonstrated below, proves consistent with the empirical evidence. Consequently, both constant-elasticity (log-log) and semi-log money demand specifications are estimated in levels, with the better-fitting relationship employed to map user costs into steady-state welfare costs via consumer surplus calculations (Section 4).

3.3 Preliminary Statistical Analysis

To justify estimation in levels and document persistence properties, this subsection reports (i) simple comovement patterns between the money-income ratio z_t and user cost i_t , and (ii) stationarity characteristics using KPSS tests.

Comovement analysis. Over the annual sample (1980–2024), z_t and i_t exhibit negative association: Pearson correlation coefficient $r = -0.275$ with two-sided p -value = 0.067. While the magnitude remains modest, this relationship is consistent with a downward-sloping money demand function.

Persistence properties. KPSS tests (employing level-stationarity null hypothesis) indicate that i_t is nonstationary in levels (KPSS p -value = 0.010), while Δi_t does not reject level-stationarity (p -value = 0.100). For z_t , the KPSS level test does not reject stationarity at the 5% significance level (p -value = 0.091), and Δz_t similarly does not reject stationarity (p -value = 0.083). Collectively, both series exhibit high persistence, with near-unit-root behaviour particularly evident for the interest rate. This finding appears plausible given documented UK regime transitions (1980s disinflation, post-2008 effective lower bound, and 2021–2024 tightening). Complete ADF test results are provided in the appendix.

Table 2: Preliminary statistical tests for z and i (annual, 1980–2024)

Test	Series	Statistic/Estimate	p -value
Pearson correlation	(z, i) (levels)	$r = -0.275$	0.067
KPSS (level-stationarity null)	z (level)	KPSS = 0.368	0.091
KPSS (level-stationarity null)	i (level)	KPSS = 1.079	0.010
KPSS (level-stationarity null)	Δz	KPSS = 0.385	0.083
KPSS (level-stationarity null)	Δi	KPSS = 0.269	0.100

Notes: KPSS employs level-stationarity null hypothesis; smaller p -values imply rejection of stationarity. ADF test summaries (drift specifications) appear in the appendix and corroborate high persistence, particularly for i_t .

3.4 Dataset Strengths and Limitations

Principal strengths. (i) *Comprehensive institutional coverage.* The 1980–2024 timespan encompasses major UK monetary policy regimes—early 1980s disinflation, ERM exit and inflation targeting adoption during the 1990s, operational independence (from 1997), post-2008 effective lower bound constraints, pandemic-related disruptions, and the 2021–2024 tightening cycle—thereby providing substantial variation in user cost i_t across both low and moderate ranges. (ii) *Framework consistency.* Defining $z_t = M_{1,t}/Y_t$ under unit income elasticity assumptions permits estimation in levels without constructing separate price deflators, while mapping directly into the welfare calculations employed below. (iii) *Data quality and transparency.* The series derive from authoritative official sources (Bank of England; ONS), contain no missing annual observations across the sample period, and employ straightforward transformations (units and logarithms) that facilitate replication.

Key limitations. (i) *Monetary aggregate definition.* M1 excludes certain close substitutes for transaction liquidity (such as interest-bearing current accounts and electronic money instruments outside M1), while its institutional coverage may have evolved with financial innovation. Such evolutionary changes can appear as shifts in money demand even when underlying preferences and technology remain stable. (ii) *Stock-flow timing conventions.* Employing end-year stock measures for M1 against annual flow measures for GDP follows standard practice but introduces measurement noise if year-end balances inadequately represent within-year usage patterns. (iii) *Structural stability concerns.* Major UK developments—including the 1986 "Big Bang" financial market reforms, the 1990–1992 ERM episode, the post-1997 monetary policy regime, the global financial crisis, pandemic-related disruptions, and recent inflation resurgence—may induce parameter instability in money demand relationships. Within an annual sample framework, such breaks prove difficult to isolate cleanly. (iv) *Unit income elasticity assumption.* The assumed proportionality between real balances and income represents a simplifying device that enables z -based estimation; deviations from unit income elasticity would embed slow-moving scale effects in z_t unrelated to user cost variations.

Interpretation and implications. The UK's sophisticated financial intermediation, widespread availability of interest-bearing current accounts, and early adoption of electronic payment systems likely attenuate the interest sensitivity of transaction balances at policy-relevant rates. The dataset reflects these institutional characteristics: z_t exhibits persistence while remaining comparatively modest by advanced economy standards, whereas i_t spans regimes that enable identification of behaviour near the effective lower bound and during tightening episodes. The limitations outlined above counsel caution in interpreting point estimates (particularly regarding user cost proxies and potential structural breaks) and motivate the reporting of both log-log and semi-log specifications in Section 4.

4 Welfare Cost Estimation: Methodology and Results

This section establishes the theoretical framework for estimating the welfare cost of anticipated inflation in the United Kingdom and presents the empirical findings derived from this analysis. The methodology builds upon established principles in monetary economics, particularly the approach developed by Lucas (2000), while adapting these methods to the specific institutional and economic context of the UK. The analysis implements two distinct money demand specifications—log-log and semi-log approaches—to provide robust estimates and assess the sensitivity of welfare cost calculations to different functional form assumptions. This dual specification strategy enables a comprehensive evaluation of how inflation affects economic welfare through its impact on optimal money holdings.

4.1 Theoretical Framework and Welfare Cost Derivation

Following Bailey (1956) and Lucas (2000), anticipated inflation is modelled as a wedge (the nominal user cost i) that induces agents to hold fewer real balances than under $i = 0$. Under unit income elasticity assumptions, the empirical object becomes the money-income ratio $z(i) \equiv m(i)$, enabling welfare to be inferred directly from the money demand schedule.

Bailey-Lucas consumer surplus framework. Let $\psi(m)$ denote the inverse demand for liquidity services. When $i > 0$, chosen real balances satisfy $m(i) < m(0)$. The welfare gain from eliminating the distortion (moving from i to 0) equals net consumer surplus:

$$W(i) = \int_{m(i)}^{m(0)} \psi(x) dx = \int_0^i m(x) dx - i m(i), \quad (1)$$

where the final identity integrates in "price space" using $\psi = m^{-1}$. Since $m(i) = z(i)$ is measured per unit of income, $W(i)$ represents an income (or GDP) share.

Two standard functional forms for $z(i)$ are estimated, with each implying a corresponding closed-form expression for $W(i)$.

Log-log (constant elasticity) specification. Under the constant elasticity assumption:

$$m(i) = A i^g, \quad g < 0, \quad (2)$$

(corresponding to $\log z = \alpha + g \log i$), the welfare function becomes:

$$W(i) = \int_0^i A x^g dx - i \cdot A i^g = \frac{A}{g+1} i^{g+1} - A i^{g+1} = \frac{-g}{g+1} A i^{g+1} > 0. \quad (3)$$

For $g \in (-1, 0)$, $W(i)$ is increasing and concave in i .

Semi-log (semi-elasticity) specification. Under the semi-log assumption:

$$m(i) = B e^{-\varepsilon i}, \quad \varepsilon > 0, \quad (4)$$

(corresponding to $\log z = \beta - \varepsilon i$), the welfare function becomes:

$$W(i) = \int_0^i B e^{-\varepsilon x} dx - i \cdot B e^{-\varepsilon i} = \frac{B}{\varepsilon} \left[1 - (1 + \varepsilon i) e^{-\varepsilon i} \right]. \quad (5)$$

For small i , $W(i) \approx \frac{B}{2} \varepsilon i^2$; curvature at higher user costs is governed by the exponential term.

Conceptual interpretation. The analytical framework treats the nominal user cost of liquidity, i , as the "price" of holding transaction balances, with the money-income ratio $z = M_1/Y$ representing the quantity demanded of liquidity services per unit of income. Two estimation approaches are employed: (i) a constant-elasticity curve (log-log), where a 1% change in i generates a $g\%$ change in z ; and (ii) a semi-log curve, where a one-percentage-point change in i produces an $\varepsilon\%$ change in z . The welfare cost at any given i equals the area under the estimated demand curve from 0 to i (the Bailey triangle). Behaviour near $i \approx 0$ proves most consequential since contemporary UK observations concentrate in this region, motivating the reporting of both functional forms as robustness checks.

4.2 Empirical Implementation and Model Performance

The estimation employs annual UK data for **1980–2024**, constructed exactly as documented in Section 3. The following specifications are fitted:

$$(\text{log-log}) \quad \log z_t = \alpha + g \log i_t + \varepsilon_t, \quad (\text{semi-log}) \quad \log z_t = \beta + \gamma i_t + u_t,$$

where g represents the interest elasticity in the log-log model and $\varepsilon \equiv -\gamma$ the semi-elasticity in the semi-log model. The sample provides complete annual coverage without imputation.

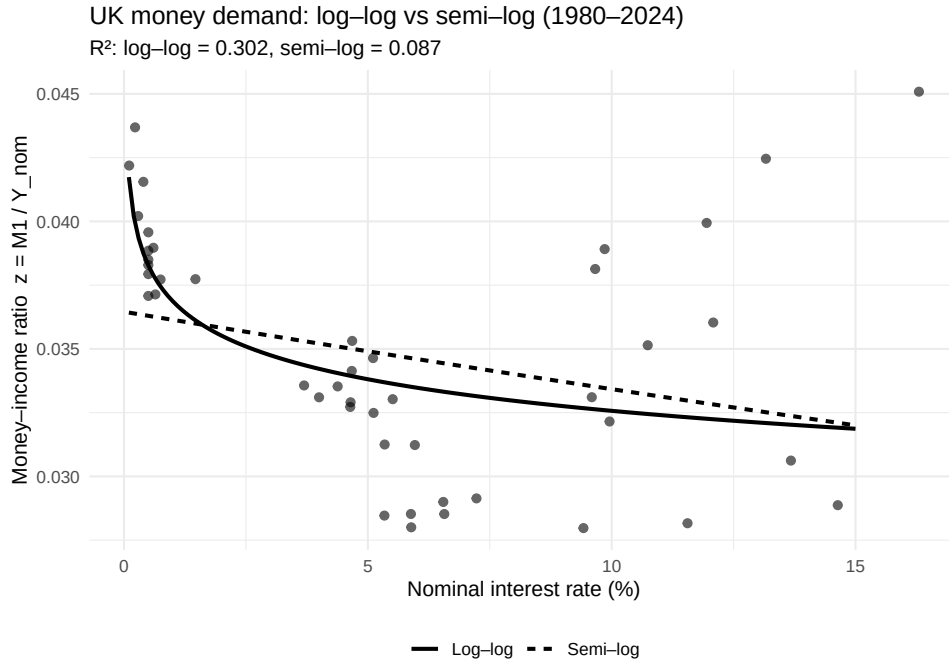


Figure 6: UK money demand, $z = M_1/Y$ versus user cost i , with log-log and semi-log fitted curves (annual, 1980–2024).

Interpreting Figure 6. Three visual characteristics prove critical for interpretation and welfare mapping: (i) the data concentrate at *low user costs* ($i \in [0, 6]\%$) with z confined to a narrow band (~ 0.03 – 0.04), while several high- i observations from the early 1980s anchor slope identification; (ii) the constant-elasticity (log-log) curve, with $A \simeq 0.0288$ and $g \simeq -0.0538$, exhibits smooth curvature and passes close to the centroid of the low- i data, whereas the semi-log curve, $z = Be^{-\varepsilon i}$ with $B \simeq 0.0365$ and $\varepsilon \simeq 0.8686$, asymptotes toward B as $i \rightarrow 0$; and (iii) since the Bailey integral exhibits greatest sensitivity near $i = 0$, these distinct curvature patterns generate different welfare implications at policy-relevant user costs—the semi-log mapping displays notably flatter behaviour.

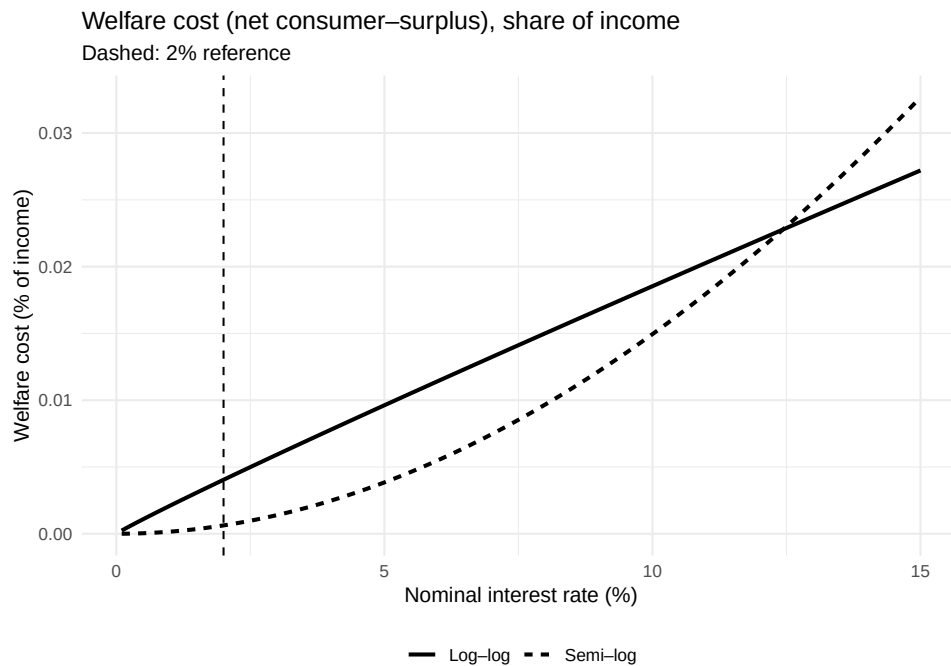
Table 3: Money demand parameter estimates employed for welfare mapping (UK, 1980–2024)

	Log-log	Semi-log
Intercept (α / β)	−3.547	−3.310
Elasticity (g / ε)	−0.0538	0.8686
Scale parameter ($A = e^\alpha / B = e^\beta$)	0.0288	0.0365
R^2	0.302	0.086
Observations	45	45

Notes: g denotes the interest elasticity in the log-log model $\log z = \alpha + g \log i$; $\varepsilon = -\gamma$ represents the semi-elasticity in the semi-log model $\log z = \beta + \gamma i$. Parameter estimates correspond to the fitted curves in Figure 6.

4.3 Welfare Cost Calculations and Magnitudes

Using equations (3) and (5), Figure 7 plots $W(i)$ against the user cost i for both specifications, employing the parameter estimates from Table 3. Since the Bailey-Lucas calculation normalises by income through $z = M_1/Y$, all reported values represent *shares of income*.

Figure 7: Welfare cost functions $W(i)$ under log-log and semi-log specifications

To provide concrete magnitudes, Table 4 reports point estimates at selected user costs. The preferred log-log specification yields modest but economically meaningful losses that increase concavely with i , while the semi-log specification exhibits flatter behaviour near the effective lower bound.

Table 4: Welfare gains from reducing user cost to zero (percent of income)

User cost (i , %)	Log-log	Semi-log
1	0.0020	0.0002
2	0.0040	0.0007
5	0.0095	0.0038
8	0.0150	0.0097
10	0.0185	0.0150

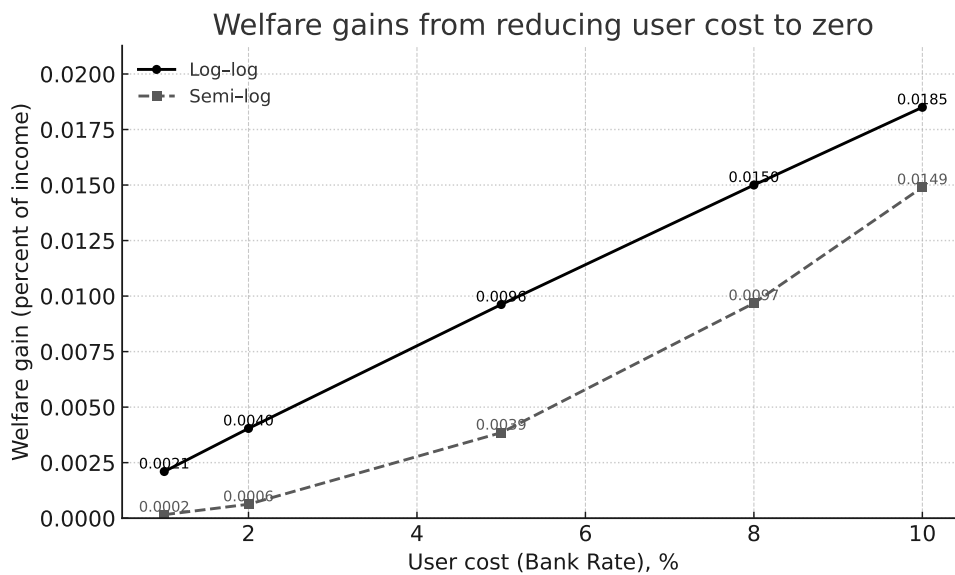


Figure 8: Welfare gains from reducing user cost to zero (UK, 1980–2024). Notes: Gains expressed as percent of income, derived from estimated money demand curves; markers indicate values at 1, 2, 5, 8, and 10% user costs.

4.4 Contextualizing UK Results Within Existing Evidence

Position within advanced economy estimates. The UK welfare cost magnitudes reported in Table 4—single-digit basis points of income at policy-relevant user costs (1–2%) and modestly higher but still small values at 8–10%—position the UK at the *lower end* of advanced economy estimates documented in Section 2.2. These findings align with post-1980 U.S. Bailey triangle results documenting small steady-state losses at low inflation (Ireland, 2009; Lucas, 2000) and with microeconomic "shopping-time" evidence suggesting limited resource costs of cash management in modern payment environments (Attanasio et al., 2002). The UK estimates thus corroborate the broader OECD pattern whereby policy-relevant anticipated inflation imposes modest deadweight losses in steady state.

Explaining the UK's modest welfare costs. The relatively flat UK money demand curve in the region where observations concentrate (Figure 6) appears plausible given institutional characteristics highlighted in Section 2.2: widespread interest-bearing current accounts, sophisticated electronic payment systems (Faster Payments, contactless transactions, mobile payments), and minimal rebalancing frictions. These features effectively reduce the user cost of holding transaction balances relative to close substitutes, thereby compressing both the estimated elasticity and the implied Bailey triangle area at any given user cost.

Functional form implications near the effective lower bound. As emphasised by Lucas (2000) and confirmed in subsequent cross-country research, welfare calculations depend critically on money demand behaviour near the origin. The side-by-side UK specifications illustrate this sensitivity: the constant-elasticity (log-log) curve fits slightly better to the centroid of low- i observations and consequently generates somewhat larger (though still modest) welfare costs $W(i)$ than the semi-log curve at policy-relevant user costs. This pattern corresponds to mixed international evidence on log-log versus semi-log functional forms summarised in Section 2.2: while both approaches can accommodate mid-range observations adequately, their curvature behaviour near $i = 0$ proves decisive for welfare cost calculations.

Comparison with emerging market economies. Section 2.2 demonstrates that identical methodological approaches generate *substantially larger* welfare losses in economies characterised by cash-intensive transactions and limited money substitutes (such as China, Turkey, India, and Brazil). The UK's modest Bailey triangles therefore align with expectations based on institutional development: sophisticated payments infrastructure implies flatter money demand, which in turn generates lower steady-state welfare costs for any given user cost.

Measurement units and cross-study comparability. Welfare costs are reported as *income shares*. While some studies employ consumption as the denominator, income and consumption shares prove similar in magnitude for advanced economies over long horizons, leaving qualitative rankings and policy conclusions unaffected. Where necessary, comparisons with Section 2.2 should be interpreted as "order-of-magnitude" benchmarks rather than precise quantitative targets.

Summary assessment. Evaluated against existing literature, the UK appears positioned near the efficiency frontier among advanced economies: *low* user costs generate *modest* steady-state efficiency losses. The stronger curvature exhibited by the log-log specification relative to the semi-log approach at very low i explains why the preferred

specification yields slightly larger (yet still modest) welfare estimates and supports the conclusion that low positive inflation targets remain consistent with near-Friedman-rule efficiency while preserving macroeconomic stabilisation capacity.

4.5 Friedman Rule Interpretation and Policy Implications

This subsection interprets the UK estimates through the theoretical lens of the Friedman rule and examines why *low* inflation targeting proves welfare-enhancing within this analytical framework, concluding with policy implications.

From theoretical optimum to practical implementation. The Friedman rule prescribes setting the nominal user cost to zero ($i = 0$) such that agents achieve liquidity satiation and the Bailey triangle disappears (cf. equations (3)–(5)). In steady state, the user cost approximates $i \approx r + \pi$ (Fisher equation). Since the long-run real interest rate r remains largely beyond monetary policy control, policy can compress i by maintaining low trend inflation π . The UK estimates quantify how closely the current monetary regime approaches this theoretical benchmark.

Mapping UK welfare costs to the Friedman rule. Under the preferred log-log specification, welfare losses at policy-relevant user costs prove *minimal*: Table 4 indicates $W(2\%) \approx 0.0040\%$ of income and $W(1\%) \approx 0.0020\%$. These represent precisely the deadweight losses avoided by progressing from positive user costs toward $i = 0$. The semi-log specification yields similarly modest values near the effective lower bound (e.g., $W(2\%) \approx 0.0007\%$), reinforcing the conclusion that the contemporary UK monetary regime maintains only minimal wedges between private and social liquidity costs.

Diminishing returns to further disinflation. Under the constant-elasticity specification (3), $W(i)$ exhibits *concavity* in i for $g \in (-1, 0)$. Marginal gains from reducing already low user costs therefore remain limited:

$$\Delta W(5\% \rightarrow 2\%) \approx 0.0095 - 0.0040 = 0.0055\% \text{ of income, } \Delta W(2\% \rightarrow 0\%) \approx 0.0040\%.$$

Conversely, maintaining elevated user costs (e.g., 8–10%) scales welfare losses upward by factors of approximately 3.8–4.6 relative to 2% (0.0150%–0.0185% versus 0.0040%). The welfare arithmetic thus favours maintaining low- i monetary regimes.

Constraints on literal Friedman rule implementation. Direct implementation of the Friedman rule (trend deflation achieving $i = 0$) encounters practical second-best constraints:

- *Effective lower bound considerations.* Modest positive π preserves scope for policy rate reductions during economic downturns; strict $i = 0$ may amplify business cycle welfare losses.
- *Fiscal policy interactions.* Eliminating seigniorage revenue necessitates higher distortionary taxation, potentially generating deadweight losses elsewhere; optimal policy requires balancing multiple tax wedges rather than eliminating only the monetary distortion.
- *Institutional and structural frictions.* Nominal rigidities, payment system characteristics, and distributional or liquidity frictions beyond the transaction margin may render slightly positive inflation attractive in general equilibrium, even when Bailey triangle calculations alone favour $i \rightarrow 0$.

Consistency with UK institutional characteristics. The UK's extensive use of interest-bearing current accounts and sophisticated electronic payment infrastructure effectively reduces the private user cost of maintaining transaction balances. This institutional context aligns with Figure 6: observations cluster at low i , the fitted log-log curve exhibits relative flatness in this region, and the resulting $W(i)$ functions displayed in Figure 7 and Figure 8 remain modest at policy-relevant values.

Policy implications. Maintaining *low* inflation keeps the user cost wedge close to the Friedman ideal, resulting in very small steady-state welfare losses from liquidity economising (e.g., $W(2\%) \approx 0.0040\%$). Given the concavity of $W(i)$, further disinflation would yield only modest steady-state gains, while positive inflation targeting preserves macroeconomic stabilisation capacity and avoids second-best fiscal costs. By this metric, the UK operates near the efficiency frontier: low inflation proves beneficial for economic welfare.

5 Conclusions

This dissertation quantified the steady-state welfare cost of anticipated inflation in the United Kingdom using the Bailey–Lucas consumer-surplus mapping applied to annual data for 1980–2024. The core mechanism is simple: a positive nominal user cost of liquidity i induces households and firms to economise on transactions balances, diverting real resources into cash-management activity. The associated deadweight loss equals the area under the money-demand curve between i and the zero-cost Friedman benchmark. Estimating money demand in levels under two canonical forms—log–log and semi-log—and mapping the fitted schedules into closed-form welfare expressions, I find that at low, policy-relevant user costs the implied welfare losses are small as a *share of income*.

5.1 Summary of findings

Three results are robust across specifications. *First*, the UK money-demand schedule is flat near the lower bound. Under the log–log form, moving from $i=2\%$ to 0% yields a welfare gain of about 0.0040% of income; under the semi-log form, the gain is roughly 0.0007%. *Second*, the welfare schedule is concave in i : marginal benefits from further disinflation shrink as $i \rightarrow 0$, while the losses associated with higher i rise more than proportionally. *Third*, at elevated user costs the absolute triangle increases several-fold—e.g., $W(8\%) \approx 0.0150\%$ and $W(10\%) \approx 0.0185\%$ under the log–log mapping—underscoring the welfare value of avoiding sustained periods of high nominal rates.

5.2 Interpretation relative to the Friedman rule

The Friedman rule recommends $i=0$ when money can be provided at negligible social cost and when lowering i carries no offsetting macro risks. In the modern UK environment, the steady-state wedge between the current low-inflation regime and the Friedman ideal is *tiny* on this metric. In a realistic, second-best setting—with stabilisation needs, ELB risks, and fiscal interactions—the efficiency case for pushing i literally to zero is weak. A positive, low inflation target preserves policy space at a very small steady-state efficiency cost.

5.3 Policy implications

- **Target design.** With user costs near zero, the incremental welfare gain from further disinflation is minute; the value of stabilisation capacity and lower-bound insurance looms larger.
- **Avoid persistence at high i .** Prolonged episodes of elevated user costs generate quickly rising losses (e.g., 8–10% vs. 2%), strengthening the case for timely, credible

disinflation when inflation drifts above target.

- **Measurement.** Because aggregate choice and user-cost proxies affect levels, continued investment in consistent monetary aggregates and retail-margin user-cost measures would improve ongoing monitoring of the welfare wedge.

5.4 Robustness and sensitivity

Using both log-log and semi-log forms brackets plausible curvatures around $i \approx 0$; the “small at low i ” conclusion is common to both. Remaining sensitivities include the choice of transactions aggregate (narrow vs. broader liquidity), the user-cost proxy (policy vs. retail rates), and structural change (financial liberalisation, payment innovation, regulatory shifts). Simple diagnostics and exclusion of extreme years point to stability of the headline results, but time-variation in elasticities remains a natural extension.

5.5 Limitations

The analysis is partial-equilibrium and abstracts from heterogeneity, risk, and general-equilibrium feedbacks between monetary and fiscal policy. It treats the marginal cost of liquidity provision as negligible, whereas balance-sheet regulation and operational frictions may make very low i costly or risky. Institutional reclassification of monetary aggregates and national-accounts revisions can introduce measurement noise that attenuates elasticities. Accordingly, the estimates should be read as steady-state benchmarks, not fine-tuned short-run prescriptions.

5.6 Directions for future research

1. **User-cost measures.** Construct Divisia-style or retail-rate user costs and re-estimate welfare schedules.
2. **Liquidity scope.** Extend the aggregate to interest-bearing transaction-like accounts and e-money to gauge sensitivity to breadth.
3. **Time variation.** Allow elasticities to vary (rolling, regime-switching, or state-dependent models) to track structural change.
4. **GE benchmarks.** Calibrate MIU/CIA/shopping-time/search models for the UK to compare partial- vs. general-equilibrium welfare.
5. **Distribution.** Study heterogeneity in effective user costs and welfare across households and SMEs given payment-tech access.
6. **Micro evidence.** Use bank- or household-level data on deposits, payments, and pass-through to identify micro foundations of the aggregate curve.

5.7 Broader relevance

Advanced economies with sophisticated payments infrastructure and credible policy frameworks tend to display interest-inelastic money demand at low i , implying small steady-state welfare losses from low, positive inflation. The welfare calculus is more salient where i is elevated or volatile, credibility is weaker, or cash dependence is high. In that international context, the UK sits toward the “small losses” end of the spectrum.

5.8 Final remarks

The Bailey–Lucas mapping translates a complex policy question into a simple metric: percent-of-income deadweight loss from anticipated inflation. For today’s UK, that metric is small at policy-range user costs. This does not trivialise inflation—unanticipated components, distributional channels, and nominal rigidities still matter—but it does indicate that, on the narrow criterion of steady-state liquidity distortion, the current low-inflation regime is close to efficient. The policy message is straightforward: maintain credibility for low, stable inflation; avoid prolonged periods of high nominal rates; and recognise that pushing all the way to the Friedman rule would yield only tiny additional efficiency gains while potentially reducing stabilisation capacity.

Acknowledgement of Generative AI Use

I acknowledge the use of **ChatGPT (GPT-5 Thinking, OpenAI, <https://chat.openai.com/>)** for two limited purposes: (i) gaining insights on drafting and structuring the report; and (ii) providing feedback or light proofreading for paragraphs without substantial mathematical content. All data analysis, empirical results, derivations, and final interpretations are my own, and all cited claims were independently verified against the original sources.

A Estimation script (R)

Listing 1: Minimal estimation script used in Section 4

```

1 #####
2 # Estimation script
3 # Input : uk_merged_data_for_lucas.csv
4 # Uses : columns `year`, `z`, `i` (i in decimal, z = M1/Y)
5 # Output: model_comparison_uk.csv
6 #         welfare_costs_comparison_uk.csv
7 #         fig_money_demand_specifications.pdf
8 #         fig_welfare_cost_functions.pdf
9 #         fig_welfare_gains_line_bw.pdf
10 #####
11
12 suppressPackageStartupMessages({
13   library(tidyverse)
14 })
15
16 # 1) Load and keep only what's needed
17 df <- read.csv("uk_merged_data_for_lucas.csv", check.names = FALSE)
18   %>%
19   transmute(
20     year = as.numeric(year),
21     z     = as.numeric(z),    # money income ratio (M1 / Y)
22     i     = as.numeric(i)    # user cost in DECIMAL (e.g. 0.02 for 2%)
23   ) %>%
24   arrange(year) %>%
25   filter(is.finite(year), is.finite(z), is.finite(i), z > 0, i > 0)
26 stopifnot(nrow(df) > 0)
27

```

```

28 # 2) Estimate money demand
29 # L o g log :  $\log z = \alpha + g * \log i$ 
30 fit_loglog <- lm(log(z) ~ log(i), data = df)
31 alpha <- unname(coef(fit_loglog)[1])
32 g      <- unname(coef(fit_loglog)[2])      # should be negative
33 A      <- exp(alpha)
34 R2_ll  <- summary(fit_loglog)$r.squared
35
36 # S e m i log :  $\log z = \beta + \gamma * i$  (semi-elasticity =  $-\gamma =$ 
    epsilon > 0)
37 fit_semilog <- lm(log(z) ~ i, data = df)
38 beta      <- unname(coef(fit_semilog)[1])
39 gamma     <- unname(coef(fit_semilog)[2])
40 epsilon   <- -gamma
41 B         <- exp(beta)
42 R2_sl     <- summary(fit_semilog)$r.squared
43
44 # Save model comparison table
45 model_tbl <- tibble(
46   Specification = c(" L o g log ", " S e m i log "),
47   Intercept      = c(alpha, beta),
48   Elasticity     = c(g, epsilon),      # g for l o g log ; epsilon for
    s e m i log
49   Scale          = c(A, B),
50   R2              = c(R2_ll, R2_sl),
51   Observations   = nrow(df)
52 )
53 write.csv(model_tbl, "model_comparison_uk.csv", row.names = FALSE)
54
55 # 3) Welfare mapping (shares of income)
56 W_loglog <- function(i, A, g) {
57   (-g / (1 + g)) * A * (i^(1 + g))
58 }
59 W_semilog <- function(i, B, eps) {
60   (B / eps) * (1 - (1 + eps * i) * exp(-eps * i))
61 }
62
63 i_grid <- seq(0.0005, 0.12, by = 0.0005)      # 0.05% to 12% user cost
64 w_df <- tibble(
65   i_pct      = 100 * i_grid,
66   W_loglog   = W_loglog(i_grid, A, g),

```

```

67   W_semilog= W_semilog(i_grid, B, epsilon)
68 )
69
70 # Selected points for table/markers
71 i_pts <- c(0.01, 0.02, 0.05, 0.08, 0.10) # 1%, 2%, 5%, 8%, 10%
72 w_pts <- tibble(
73   user_cost_pct = 100 * i_pts,
74   loglog_percent = 100 * W_loglog(i_pts, A, g),
75   semilog_percent = 100 * W_semilog(i_pts, B, epsilon)
76 )
77 write.csv(w_pts, "welfare_costs_comparison_uk.csv", row.names = FALSE
78 )
79 # 4) Figures (black & white, different linetypes/shapes)
80
81 # 4a) Money demand: scatter + fitted curves
82 pred_df <- tibble(
83   i = i_grid,
84   z_loglog = exp(alpha + g * log(i_grid)),
85   z_semilog = exp(beta + gamma * i_grid),
86   i_pct = 100 * i_grid
87 )
88
89 p_md <- ggplot() +
90   geom_point(data = df, aes(x = 100 * i, y = z),
91             shape = 16, size = 1.6, color = "black", alpha = 0.75) +
92   geom_line(data = pred_df, aes(x = i_pct, y = z_loglog, linetype = "
93     L o g l o g "),
94            linewidth = 1, color = "black") +
95   geom_line(data = pred_df, aes(x = i_pct, y = z_semilog, linetype =
96     " S e m i l o g "),
97            linewidth = 1, color = "black") +
98   scale_linetype_manual(values = c(" L o g l o g " = "solid", " S e m i l o g
99     " = "dashed"), name = "") +
100   labs(x = "User cost i (percent)", y = expression(z == M[1]/Y),
101         title = "UK money demand: z versus user cost i (annual, 1980
102           2024 )") +
103   theme_minimal(base_size = 11) +
104   theme(legend.position = "bottom", panel.grid.minor = element_blank
105     ())

```

```

102 ggsave("fig_money_demand_specifications.pdf", p_md, width = 7.2,
      height = 4.4)
103
104 # 4b) Welfare cost functions W(i) (percent of income)
105 p_w <- ggplot(w_df, aes(x = i_pct)) +
106   geom_line(aes(y = 100 * W_loglog, linetype = "L o g log"),
107             linewidth = 1, color = "black") +
108   geom_line(aes(y = 100 * W_semi log, linetype = "S e m i log"),
109             linewidth = 1, color = "black") +
110   scale_linetype_manual(values = c("L o g log" = "solid", "S e m i log"
111     " = "dashed"), name = "") +
112   labs(x = "User cost i (percent)", y = "W(i), percent of income",
113        title = "Welfare cost functions under l o g log and s e m i log
114        mappings") +
115   theme_minimal(base_size = 11) +
116   theme(legend.position = "bottom", panel.grid.minor = element_blank
117         ())
118
119 ggsave("fig_welfare_cost_functions.pdf", p_w, width = 7.2, height =
120       4.4)
121
122 # 4c) Welfare gains at selected user costs (line with markers)
123 p_line <- w_pts %>%
124   pivot_longer(cols = c(loglog_percent, semilog_percent),
125               names_to = "model", values_to = "W_pct") %>%
126   mutate(model = recode(model,
127                         loglog_percent = "L o g log ",
128                         semilog_percent = "S e m i log ")) %>%
129   ggplot(aes(x = user_cost_pct, y = W_pct, linetype = model, shape =
130     model)) +
131   geom_line(color = "black", linewidth = 0.9) +
132   geom_point(color = "black", size = 2) +
133   scale_linetype_manual(values = c("L o g log" = "solid", "S e m i log"
134     " = "dashed"), name = "") +
135   scale_shape_manual(values = c("L o g log" = 16, "S e m i log" = 1),
136                      name = "") +
137   labs(x = "User cost i (percent)", y = "W(i), percent of income",
138        title = "Welfare gains from reducing user cost to zero (
139        selected points)") +
140   theme_minimal(base_size = 11) +

```

```
133   theme(legend.position = "bottom", panel.grid.minor = element_blank  
134         ())  
135 ggsave("fig_welfare_gains_line_bw.pdf", p_line, width = 7.2, height =  
        4.4)
```

B OLS regression outputs

B.1 Full console summaries

The full `summary(lm())` outputs for reproducibility.

Log-log: $\log z = \alpha + g \log i$

```
Call:
lm(formula = log_z ~ log_i, data = uk_ll)

Residuals:
    Min       1Q   Median       3Q      Max
-0.17958 -0.04493 -0.00891  0.03341  0.35145

Coefficients:
              Estimate Std. Error t value      Pr(>|t|)
(Intercept) -3.54829     0.04663  -76.100 < 0.0000000000000002 ***
log_i        -0.05384     0.01247   -4.316    0.0000915 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1143 on 43 degrees of freedom
Multiple R-squared:  0.3023,    Adjusted R-squared:  0.2861
F-statistic: 18.63 on 1 and 43 DF,  p-value: 0.00009148
```

Semi-log: $\log z = \beta + \gamma i$

Call:

```
lm(formula = log_z ~ i, data = uk)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.21257	-0.07071	-0.00672	0.07184	0.35408

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3.31168	0.03124	-105.993	<0.0000000000000002 ***
i	-0.86858	0.43034	-2.018	0.0498 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1308 on 43 degrees of freedom

Multiple R-squared: 0.08654, Adjusted R-squared: 0.0653

F-statistic: 4.074 on 1 and 43 DF, p-value: 0.04982

C Supplementary tables

C.1 Main approaches to measuring the welfare cost of anticipated inflation

Table 5: Main approaches to measuring the welfare cost of anticipated inflation

Approach	Core idea / ingredients	Strengths / caveats
Bailey consumer-surplus	Estimate $m(i)$ and integrate $\int_0^i m(x) dx - i m(i)$ (Bailey, 1956; Lucas, 2000).	Transparent; hinges on functional form (log-log vs. semi-log), money aggregate, and rate proxy.
Money-in-utility (MIU)	Utility includes real balances; map $i \rightarrow m(i)$; compute consumption-equivalent changes.	Microfoundation for liquidity; calibration-sensitive.
Cash-in-advance (CIA)	Transactions require cash; higher i tightens CIA constraint.	Highlights transactions role; depends on CIA share and calibration.
Shopping-time / cash-credit	Time/resources to obtain liquidity fall with money; inflation raises shopping time.	Intuitive resource-cost channel; requires technology calibration.
Search-theoretic exchange	Lagos-Wright; pricing (competitive vs. bargaining) changes surplus split (Boel & Camera, 2011; Lagos & Wright, 2005).	Structural $m(i)$; with bargaining, triangle often <i>understates</i> costs; fit varies by country.
Heterogeneous-agent GE	Inflation interacts with portfolio/liquidity heterogeneity; compute consumption-equivalent losses.	Rich distributional channels; model-dependent; not directly comparable to triangle.

C.2 Welfare-cost benchmarks: advanced economies

Table 6: Welfare-cost benchmarks — advanced economies

Country/Group	Method	Scenario / reported magnitude (source)
United States	Consumer-surplus (Bailey–Lucas)	1900–1994: reducing steady inflation 10% $\rightarrow$ 0% yields gain <i>slightly</i> $< 1\%$ of income (Lucas, 2000).
United States	Consumer-surplus (semi-log, post-1980)	Post-1980: at 2% inflation ≈ 0.036 – 0.038% of income; at 10% ≈ 0.224 – 0.232% (Ireland, 2009).
United States	CIA (general equilibrium)	Benchmark RBC–CIA: sustained 10% inflation costs ≈ 0.11 – 0.40% of GNP (Cooley & Hansen, 1989).
United States (calibration)	Search-theoretic (Lagos–Wright)	Liquidity in decentralised trade: 10% $\rightarrow$ 0% disinflation worth ≈ 3 – 5% of consumption (Craig & Rocheteau, 2006; Lagos & Wright, 2005).
Italy	Shopping-time / inventory (micro)	Household micro (cash stocks, trips, ATMs): welfare cost in policy-relevant range $\lesssim 0.1\%$ of consumption (Attanasio et al., 2002).
OECD (23 members)	Search-theoretic (price-taking vs. bargaining)	Price-taking: many $< 1\%$; Japan $\sim 0.1\%$; Australia $> 2\%$. With bargaining frictions: up to $\sim 6\%$ (Boel & Camera, 2011).
Canada vs. U.S.	Consumer-surplus (long-horizon)	1948–2001 (Fisher–Seater elasticity). U.S.: 14% $\rightarrow$ 3% gain $\approx 0.45\%$; 3% $\rightarrow$ 0% $\approx 0.18\%$. Canada: 14% $\rightarrow$ 3% $\approx 0.35\%$; 3% $\rightarrow$ 0% $\approx 0.15\%$ (Serletis & Yavari, 2004).

Notes: Magnitudes are consumption/income-equivalents where stated. Consumer-surplus figures depend on money-demand form and user-cost proxy; CIA/search models incorporate GE channels.

C.3 Welfare-cost benchmarks: emerging and developing economies

Table 7: Welfare-cost benchmarks — emerging and developing economies

Country/Group	Method	Scenario / reported magnitude (source)
China	Consumer-surplus (log-log & semi-log); MIU/CIA (GE)	CS (log-log): at 10% $\approx$ 11.25% of consumption (8%: 9.41%; 5%: 6.46%). CS (semi-log): 9.69% at 10%. CIA GE: 0.912% at 10%. 1996 annual loss $\approx$ 12.99% of consumption ($\sim$ 4.38% of GDP); 2005 $\approx$ 0.87% of GDP (Chen & Ma, 2007).
Brazil	Consumer-surplus (cointegration/-DOLS); time-varying demand	Post-Real Plan (1995–2011): 4.5% $\approx$ 0.15–0.20%; 10% $\approx$ 0.3–0.45%. Time-varying cointegration/Kalman confirm similar magnitudes (Campos & Cysne, 2019).
South Africa	Consumer-surplus (Johansen vs. Fisher-Seater)	Target band 3–6%: Johansen 0.34–0.67% of GDP; long-horizon 0.16–0.36% (Gupta & Uwilingiye, 2008, 2009).
Turkey	Consumer-surplus (log-log)	0%→10%: welfare cost 0.52–0.54% of GDP (M1, 1970–2013) (Tümtürk, 2017).
India	Consumer-surplus + compensating variation (GE)	10% inflation: $\sim$ 0.53% of GDP (1996–2015) (Shah et al., 2019).
Colombia	MIU (Sidrauski-type)	10% inflation: 1.37% of GNP; 20%: $\sim$ 2.4% of GDP (López, 2000).

Notes: Reported magnitudes are consumption/income-equivalents where stated. Cross-study differences reflect money aggregate/opportunity-cost definitions and long-run identification.

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